



MINT: Machine Intelligence for the Nest Thermostat

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Background

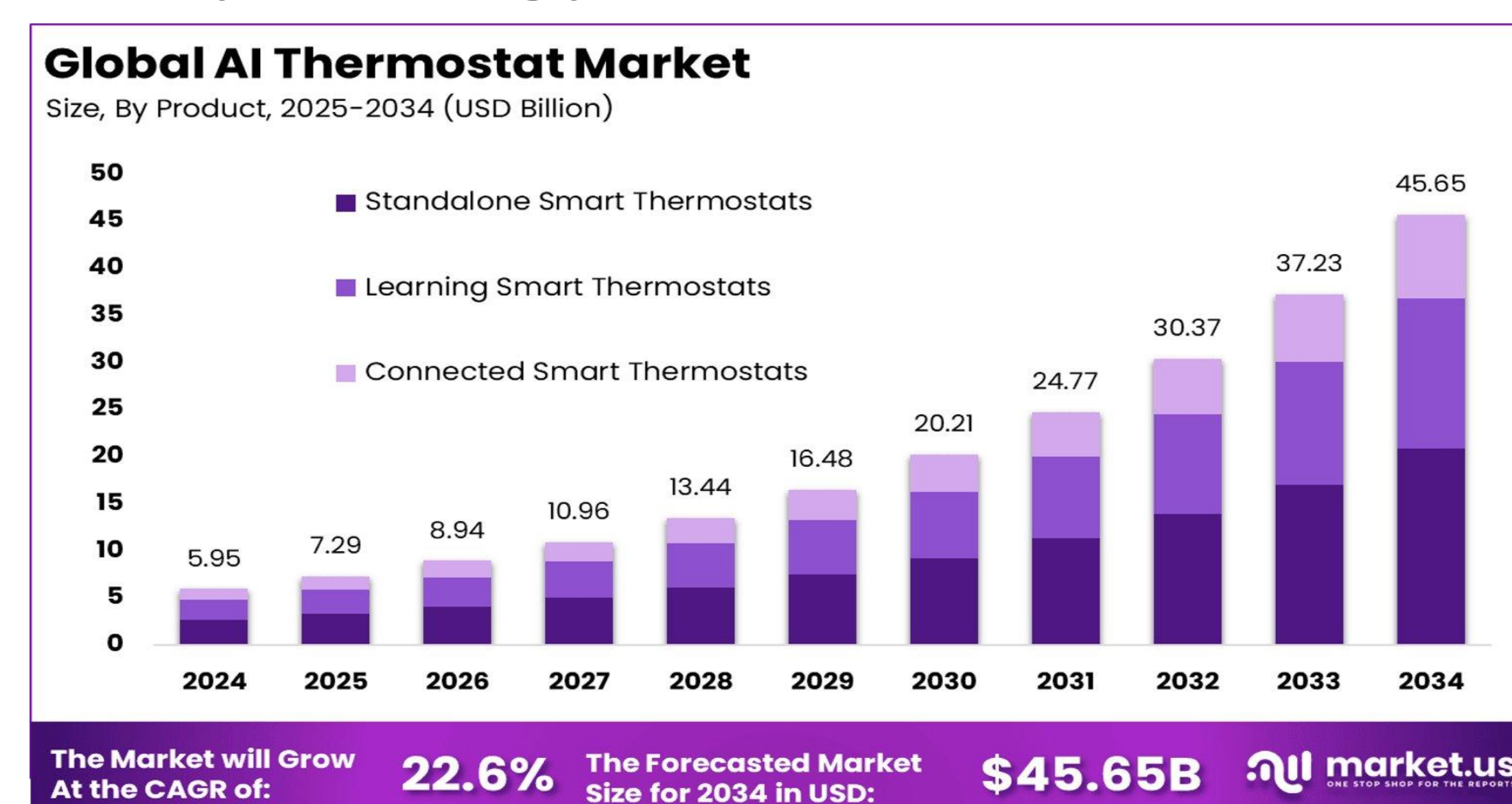
- Most “smart” thermostats learn **user schedules**, but ignore factors like insulation and system response time
- This leads to **inefficient energy use and inconsistent comfort**
- Our approach uses previously collected **weather and indoor sensor data** to model comfort and system behavior for smarter control

Project Goals

- Develop a **machine learning model** that predicts comfort and recommends energy-optimal set temperatures.
- Design and build a **thermal chamber** to validate our machine learning model

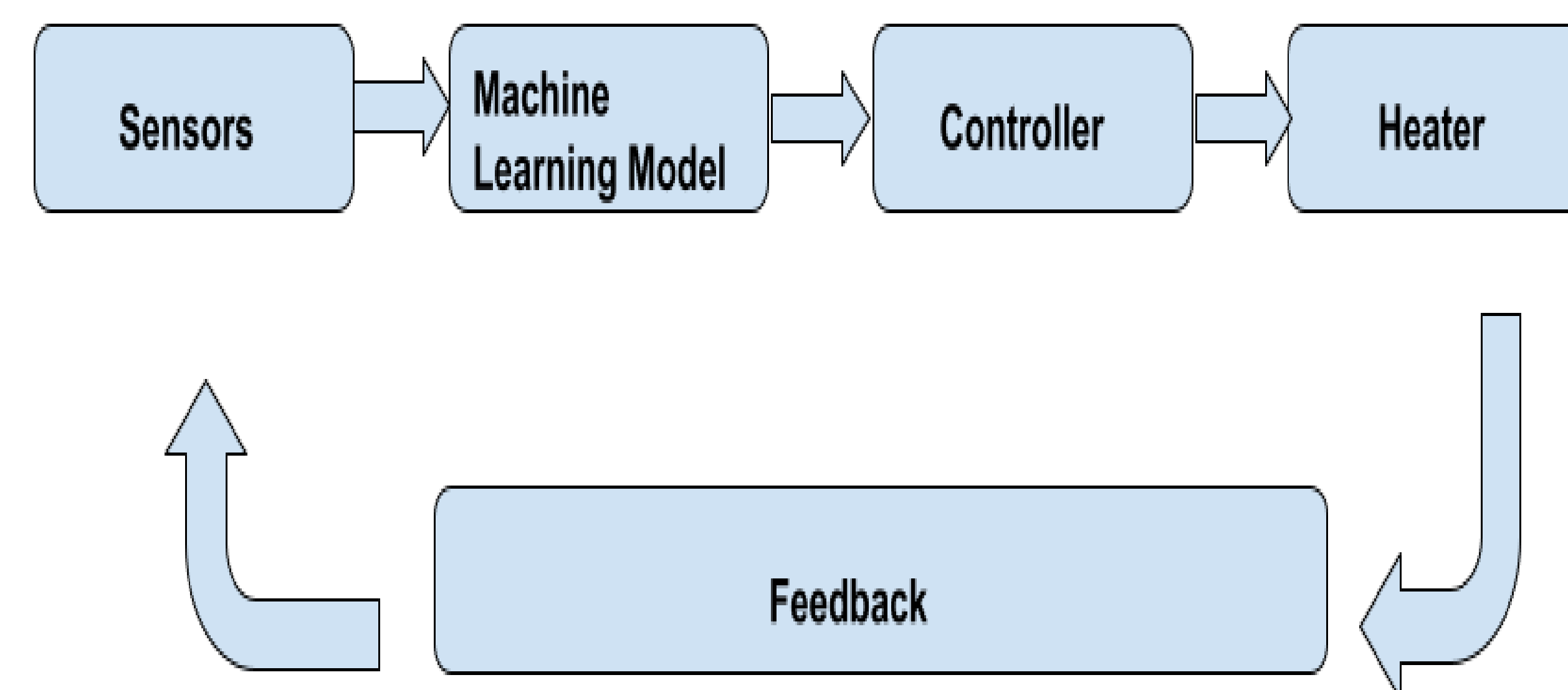
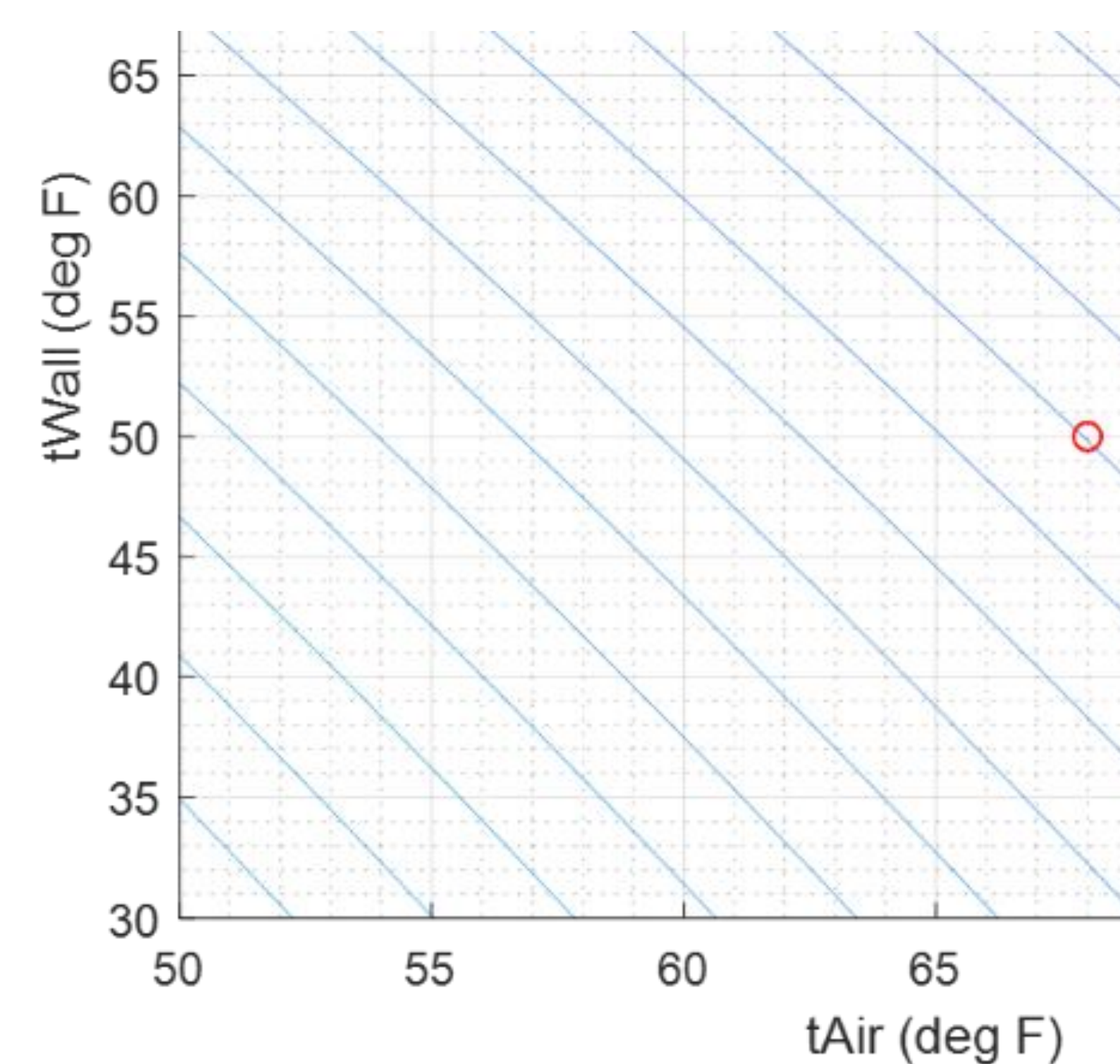
Stakeholders

Smart thermostat users (including Dr. Gray), energy conscious households



Solution Concept

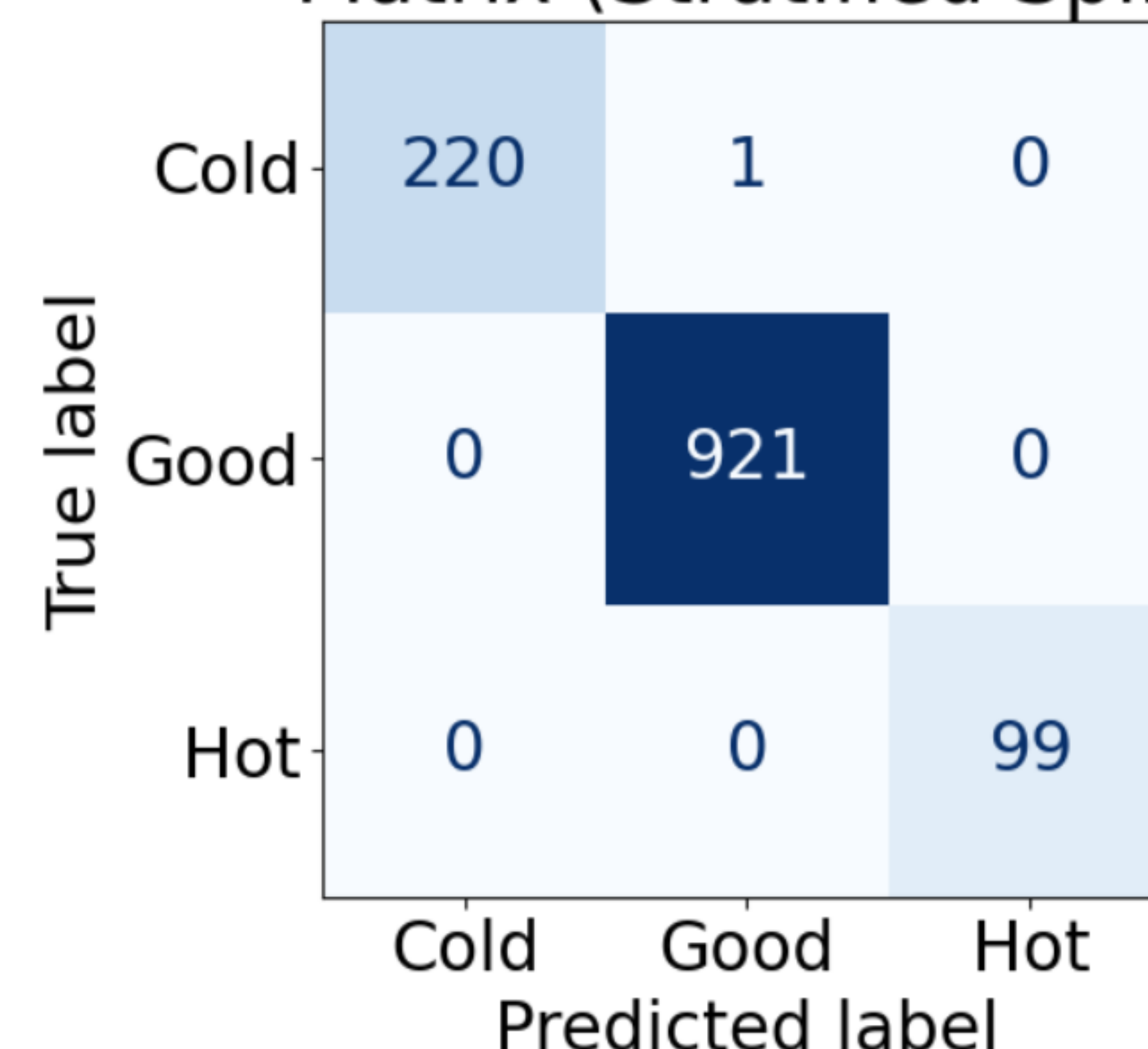
- Our solution uses a Random Forest ML model to **predict the optimal temperature setpoint**
- The ML model incorporates **physical system behavior** using a steady-state mathematical model that we built
- Our math model uses indoor air temperature with wall temperature to model user comfort
- The system will be validated using a **thermal chamber test environment**



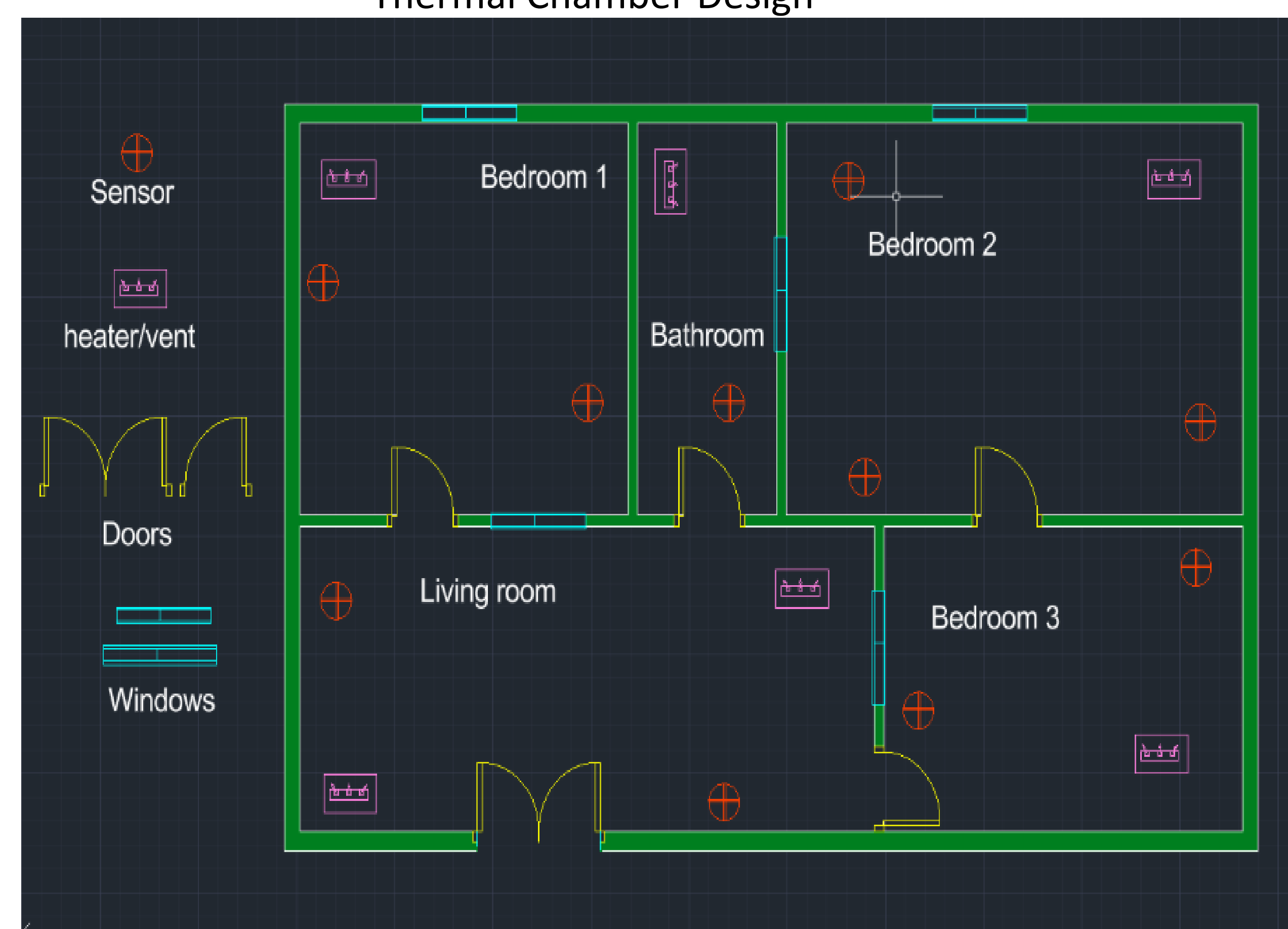
Impact

- Transformed raw thermostat and environmental data into a **practical predictive system** for comfort-aware, energy-conscious HVAC control.
- Created a machine learning pipeline from raw sensor CSV -> physics-based features -> trained RF model.
- In the long term, our work could help improve energy efficiency in residential HVAC, which would save money and conserve resources.

Random Forest Confusion Matrix (Stratified Split)



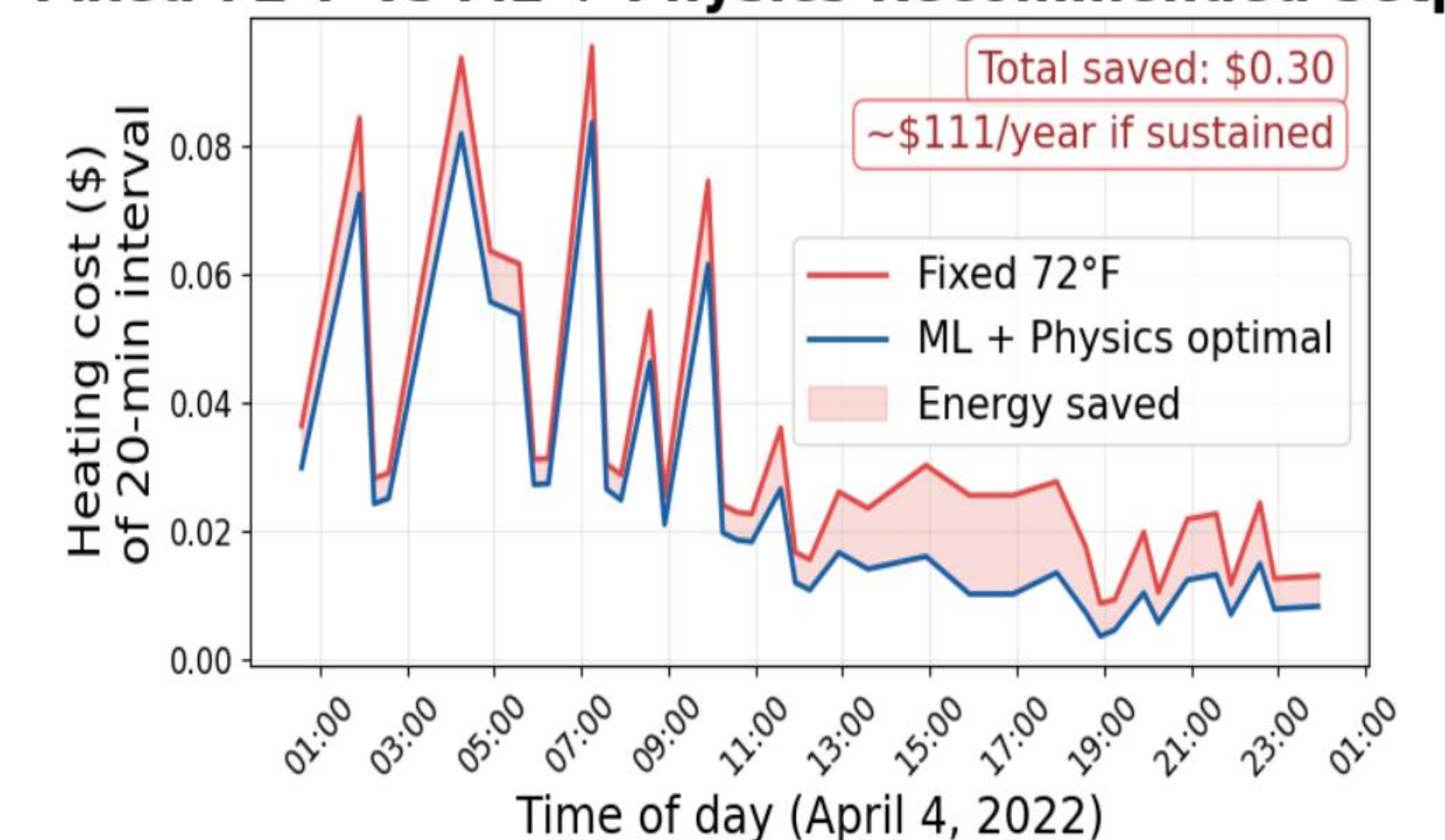
Thermal Chamber Design



Results & Reflection

- Our ML model predicts user comfort very accurately (near 100% accuracy)
- After predicting user comfort, our model takes energy efficiency into account and predicts the optimal set temperature
- We have also completed the design of the thermal chamber that will allow future teams to move this work from simulation into real-world validation and system testing.

Heating Cost Throughout the Day
Fixed 72°F vs ML + Physics Recommended Setpoint



Future Plans

Next semester:

- Build the thermal chamber** using our plans and empirically test the ML model using the chamber
- Make further improvements to our ML model and steady state model using results from testing